

1 Mathematical Formulation

1.1 Index Sets

- $K = \{1, 2, \dots, n_K\}$ as the set of all orders
- $K^0 = K \cup \{0\}$
- $C = \{1, 2, \dots, n_C\}$ as the set of all cars

1.2 Parameters

- R_k : Sales revenue of accepting order k for all $k \in K$
- S_k^{Start} : Pick up station of accepting order k for all $k \in K$
- S_k^{End} : Pick up station of accepting order k for all $k \in K$
- S_c^0 : The initial station for the car c for all $c \in C$
- B : Maximum total moving time allowed (in minutes).
- L_c : The designated level of car c . for all $c \in C$
- Req_k : The required car level for order k for all $k \in K$
- T_k^S : The start time for order k for all $k \in K$
- T_k^E : The end time for order k for all $k \in K$
- $T_{j,k}^{Trans}$: The transport time from order j to order k for all $j, k \in K$
- $T_{0,j}^{Trans}$: The transport time from the initial station to order j for all $j \in K$

1.3 Decision Variables

- $x_k \in \{0, 1\}$: 1 if order k is accepted, 0 otherwise.
- $y_{c,k} \in \{0, 1\}$: 1 if car c is assigned to execute order k , 0 otherwise.
- $u_k \in \{0, 1\}$: 1 if a free upgrade is provided for order k , 0 otherwise.
- $z_{c,i,j} \in \{0, 1\}$: 1 if car c executes order j immediately after finishing order i , 0 otherwise.

1.4 Objective Function

The objective is to maximize the total profit, which is the total sales revenue minus the total compensation for rejected orders:

$$\text{Maximize} \quad \sum_{k \in K} R_k x_k - \sum_{k \in K} 2R_k (1 - x_k) \quad (1)$$

1.5 Constraints

$$\sum_{c \in C} y_{c,k} = x_k \quad \forall k \in K \quad (2)$$

$$\sum_{c \in C} L_c y_{c,k} = Req_k x_k + u_k \quad \forall k \in K \quad (3)$$

$$\sum_{j \in K^0, j \neq k} z_{c,j,k} = y_{c,k} \quad \forall k \in K \quad \forall c \in C \quad (4)$$

$$\sum_{k \in K, j \neq k} z_{c,j,k} \leq y_{c,j} \quad \forall j \in K \quad \forall c \in C \quad (5)$$

$$\sum_{j \in K} z_{c0j} \leq 1 \quad \forall c \in C \quad (6)$$

$$T_k^S - T_j^E + V(1 - z_{c,j,k}) \geq 9 + T_{j,k}^{Trans} \quad \forall c \in C \quad j \neq k \quad \forall j \in K \quad \forall k \in K \quad (7)$$

$$T_k^S + V(1 - z_{c,0,k}) \geq 1 + T_{0,k}^{trans} \quad S_c^0 \neq S_k^{start} \quad \forall c \in C \quad \forall k \in K \quad (8)$$

$$T_k^S + V(1 - z_{c,0,k}) \geq T_{0,k}^{trans} \quad S_c^0 = S_k^{start} \quad \forall c \in C \quad \forall k \in K \quad (9)$$

$$\sum_{c \in C} \sum_{j \in K^0, j \neq k} \sum_{k \in K} T_{j,k}^{Trans} z_{c,j,k} \leq B \quad (10)$$

- **Order Assignment Constraint - (2):** This ensures that each order k is assigned to exactly one vehicle c if the order is accepted ($x_k = 1$). If the order is rejected ($x_k = 0$), no vehicle is assigned to it.
- **Vehicle Level and Upgrade Constraint - (3):** This constraint ensures that the level of the assigned vehicle L_c meets or exceeds the required level of the order Req_k . The binary or integer variable u_k captures the "upgrade" difference. By restricting the domain of u_k (e.g., $u_k \in \{0, 1\}$), the model can enforce a policy where a car can be upgraded by at most one level.
- **Network Flow Conservation (In-flow) - (4):** For each vehicle c , this constraint ensures that if a car is assigned to order k ($y_{c,k} = 1$), it must arrive from a preceding node j , which could either be the initial depot (node 0) or a previously completed order.
- **Network Flow Conservation (Out-flow) - (5):** This guarantees that after completing order j , a vehicle c may either move to a subsequent order k or cease its operations for the planning horizon. This maintains the continuity of the vehicle's path.
- **Initial Depot Departure Constraint - (6):** This limits each vehicle c to depart from its starting location (depot 0) at most once during the entire planning horizon.
- **Inter-Order Time Window Constraint - (7):** This enforces the temporal feasibility between two consecutive orders j and k . If vehicle c transitions from j to k ($z_{c,j,k} = 1$), the pick-up time T_k^S must be late enough to account for the return time of the previous order T_j^E , the mandatory cleaning duration (e.g., 240 minutes), a 30-minute buffer for readiness (totaling 270 minutes or 9 units), and the travel time $T_{j,k}^{Trans}$ between stations.

- **First Task Time Constraint (Relocation required) - (8):** When a vehicle c starts its first task at a station different from its initial depot ($S_c^0 \neq S_k^{Start}$), it must account for the travel time $T_{0,k}^{trans}$ and a 30-minute (1 unit) readiness buffer before the pick-up time T_k^S .
- **First Task Time Constraint (Immediate pick-up) - (9):** If the vehicle is already located at the pick-up station ($S_c^0 = S_k^{Start}$), the 30-minute readiness buffer is waived based on the assumption that all cars are "ready right away" at the start of the horizon. The car only needs to satisfy the travel time (which is zero in this case).
- **Total Relocation Budget Constraint - (10):** This constraint ensures that the cumulative travel time of all vehicles—including initial deployments from depots and transitions between orders—does not exceed the total allowable moving budget B .